Early-Stopping Self-Consistency

Is the saving still a gain at a matched budget on a small open-weight model?

Roll No. U23AI042
Syllabus unit: LLMs as reasoning engines (chain-of-thought and self-consistency)

Three papers, one narrow problem

How many samples does self-consistency really need?

PRIMARY

Li et al., ICLR 2024

Escape Sky-high Cost: Early-stopping Self-Consistency for Multi-step Reasoning

Stops sampling when a window of answers is unanimous. Public code. Six benchmarks, three models.

CONTEXT

Wang et al., ICLR 2023

Self-Consistency Improves Chain of Thought Reasoning in Language Models

The method being made cheaper: sample many solutions, take the majority answer.

CONTEXT

Aggarwal et al., EMNLP 2023

Let's Sample Step by Step: Adaptive-Consistency

The earlier adaptive rule: stop when a Beta posterior passes 0.95.

Self-consistency works, and it is expensive

Chain of thought, greedy: one solution, one answer. An early slip is never recovered.

Self-consistency (SC): sample L solutions at temperature 0.7 and return the most frequent answer.

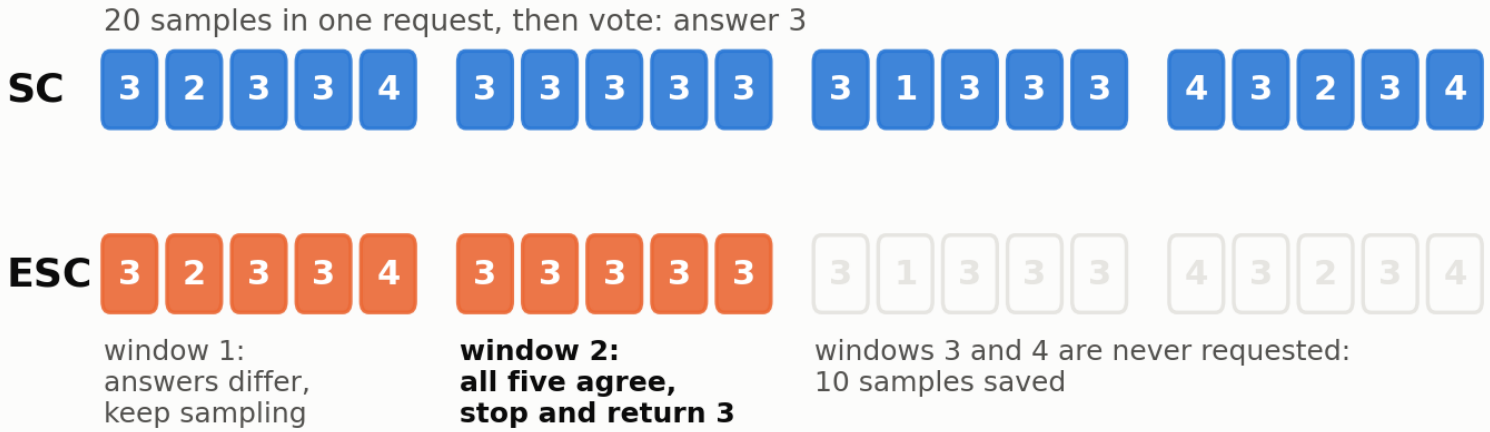
The price: cost grows in step with L, and L is the same for an easy question as for a hard one.

about $2,000

SC with 64 samples over the MATH benchmark on GPT-4
(Li et al., Sec. 1)

Most questions do not need all L samples. Which ones do?

ESC: stop when one window agrees with itself



Loop: draw w samples, check if all w answers are equal.

Equal: stop, return that answer.

Not equal: keep them, draw the next window.

Budget spent: vote over all L, exactly as SC.

No threshold to tune. Adaptive-Consistency checks a Beta posterior against 0.95 after every single sample.

Two supports: a z-test bound that an early stop rarely disagrees with full SC (Sec. 2.3), and a control scheme that predicts cost and accuracy for a choice of (w, L) (Sec. 2.4).

What the paper tested and what it reported

Setting	Value
Benchmarks	MATH, GSM8K, CSQA, StrategyQA, Last Letters, Coin Flip
Models	GPT-4, GPT-3.5-Turbo, Llama-2 7b
Prompts	few-shot chain of thought (Wei et al.)
Decoding	temperature 0.7 (0.5 for MATH)
ESC settings	w = 5, L = 40 (8 and 64 for MATH)
Runs	average of 10, variance omitted

GSM8K, GPT-4 (Table 1)

89.29%

the same accuracy as SC with 40 samples

7.98 samples

per question instead of 40

Reported saving in samples: GSM8K 80%, StrategyQA 77%, CSQA 79%, Coin Flip 84%, Last Letters 67%, MATH 34% (abstract).

Gaps 1 and 2: the comparison that matters is unmeasured

EVALUATION

The matched-budget gain is smaller than the noise nobody reported

Table 1: ESC beats SC at the same sample count by 0.22, 0.85 and 0.22 points on GSM8K.

Sec. 3.2: "averaged over 10 runs and omit variance".

0.22 points of 1,319 questions is three questions.

COST AND LATENCY

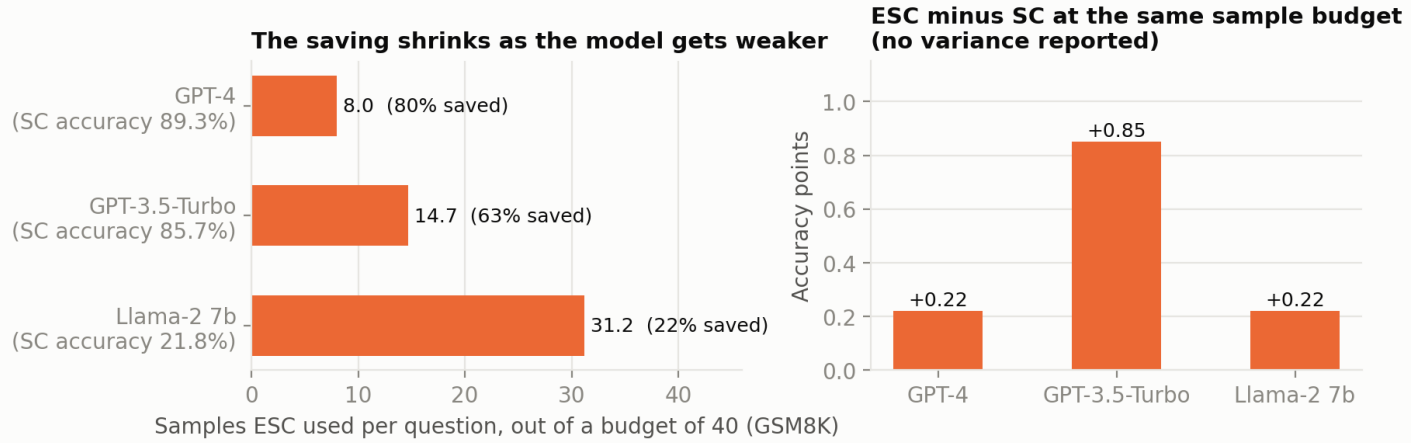
Cost is counted in samples and dollars, never in time

Tables 1 and 7: samples, tokens, dollars. No wall-clock figure.

SC asks for L samples in one parallel request. Algorithm 1 waits for each window: up to L/w = 8 round trips.

Llama-2 7b on GSM8K: 22% of samples saved, about six round trips.

Gaps 3 and 4: weak models and wrong but confident answers



GSM8K numbers from Table 1 of the paper, plotted by me.

GENERALITY

Saving falls 80% to 22% as the model weakens.

Only open-weight model: Llama-2 7b at 21.75%. Nothing in the middle, nothing laptop-sized.

ASSUMPTION

The bound is about agreeing with SC, not about being right. Table 5 reports only the overlap of ESC and SC answers; no figure for how often a stop lands on a wrong answer.

It is quoted for w = 8 (at most 0.002). Five of six benchmarks use w = 5, where z = sqrt(5) gives about 0.013.

One objective, taken from Gap 1

To evaluate early-stopping self-consistency (window 5, at most 20 samples) for arithmetic word problems on a fixed 50-question subset of GSM8K test (1,319), with Qwen2.5-1.5B-Instruct run locally, compared against fixed-size self-consistency at a matched sample budget, measured by exact-match accuracy and by samples, tokens, rounds and latency per question, as mean and standard deviation over three seeded runs.

Why this one

Repeats the paper's matched-budget row, with the variance attached.

A 1.5B model sits in the regime Gap 3 says is missing.

Latency and stop-correctness (Gaps 2, 4) come from the same traces.

Budget

20 samples x 50 questions x 3 runs = 3,000 generations.

About 0.3M completion and 0.25M prompt tokens.

Laptop CPU, llama.cpp, three to four hours, zero cost.

Plan for Phase 2

Baselines first: greedy CoT and SC at k = 1 to 20.

Then ESC on the same samples, so only the stopping rule differs.

Ablation: random stopping with ESC's budgets.